

NUTRITION AND DIETETICS

1. A 25-year-old adult has an Estimated Energy Requirement (EER) of 2,496 kcal per day. According to the Acceptable Macronutrient Distribution Range (AMDR), what is the appropriate daily carbohydrate intake range in grams for this individual?
 - a. 224.4 - 311.2 grams per day
 - b. 280.8 - 405.6 grams per day
 - c. 312.0 - 432.5 grams per day
 - d. 187.2 - 260.0 grams per day

The correct answer is: 280.8 - 405.6 grams per day

2. A 39-year-old woman follows a habitual diet consisting mainly of refined carbohydrates, sugar-sweetened beverages, pastries made with hydrogenated oils, and very little intake of fruits, vegetables, or unsaturated fats. Laboratory analysis shows that her blood cholesterol is elevated, and she has particularly low levels of the lipoprotein responsible for transporting cholesterol away from tissues and back to the liver. Which lipoprotein's function has most likely been inhibited by her dietary pattern?
 - a. VLDL
 - b. HDL
 - c. Chylomicrons
 - d. LDL

The correct answer is: HDL

3. A 52-year-old man has a habitual diet high in deep-fried fast foods, margarine made with hydrogenated oils, and commercially baked goods. Laboratory results show elevated total cholesterol and triglycerides. His physician explains that his diet has impaired the lipoprotein pathway responsible for removing cholesterol from peripheral tissues and returning it to the liver, thereby worsening his cholesterol profile. Which lipoprotein is most directly affected by this dietary pattern? A 52-year-old man has a habitual diet high in deep-fried fast foods, margarine made with hydrogenated oils, and commercially baked goods. Laboratory results show elevated total cholesterol and triglycerides. His physician explains that his diet has impaired the lipoprotein pathway responsible for removing cholesterol from peripheral tissues and returning it to the liver, thereby worsening his cholesterol profile. Which lipoprotein is most directly affected by this dietary pattern?
 - a. Chylomicrons
 - b. LDL
 - c. VLDL
 - d. HDL

The correct answer is: HDL

4. When a person consumes a carbohydrate-rich meal, glucose is absorbed into the bloodstream and transported to various tissues. To prevent dangerously high blood glucose levels, the body converts much of this glucose into glycogen for storage. This storage form of carbohydrate is strategically placed where it can either regulate blood

glucose availability or provide rapid energy during physical activity. In which tissues is glycogen most abundantly stored in the human body?

- a. Liver and skeletal muscle
- b. Small intestine and stomach
- c. Brain and pancreas
- d. Adipose tissue and kidneys

The correct answer is: Liver and skeletal muscle

5. According to Ghana's Food-Based Dietary Guidelines (FBDG), which food group contributes the largest proportion by weight to a healthy diet?
- a. Vegetables
 - b. Animal-source foods
 - c. Fruits
 - d. Cereals, roots, and tubers

The correct answer is: Cereals, roots, and tubers

6. According to the Acceptable Macronutrient Distribution Range (AMDR) guideline, what is the recommended average range of total fat intake per day for an adult male consuming a 2,500 kcal diet?
- a. 56 – 97 g
 - b. 44 – 78 g
 - c. 80 – 120 g
 - d. 30 – 60 g

The correct answer is: 56 – 97 g

7. After dietary lipids are digested and absorbed in the small intestine, they must be transported through the body's water-based environment. Which of the following best describes the structure that carries most of these lipids away from the intestinal cells and the route it takes?
- a. Fatty acids are directly absorbed into the bloodstream and travel bound to plasma proteins straight to the liver.
 - b. Cholesterol esters form VLDL particles in the intestinal wall that travel through the lymphatic system directly to adipose tissue for storage.
 - c. Phospholipids form micelles that enter capillaries in the intestine and transport dietary fats to tissues through the hepatic portal vein.
 - d. Triacylglycerols, cholesterol, and phospholipids combine with proteins to form chylomicrons, which enter the lymphatic system and are released into the bloodstream via the jugular vein.

The correct answer is: Triacylglycerols, cholesterol, and phospholipids combine with proteins to form chylomicrons, which enter the lymphatic system and are released into the bloodstream via the jugular vein.

8. Ama eats a large bowl of beans and vegetables. Although she cannot digest most of the fibre in these foods, certain microorganisms in her body help break some of it down. Where in the digestive system does this mainly occur?
- Large intestine
 - Stomach
 - Small intestine
 - Mouth

The correct answer is: Large intestine

9. Consider the following dietary assessment methods: 24-hour recall, estimated food record, weighed food record, dietary history, and food frequency questionnaire. Among these, which quantitative method(s) is/are most prone to memory error, because the respondent must depend on remembering rather than recording intake contemporaneously?
- 24-hour recall and estimated food record
 - Weighed food record only
 - Estimated food record only
 - 24-hour recall only

The correct answer is: 24-hour recall only

10. Disaccharides in the human diet are formed when two monosaccharides join through glycosidic bonds. Glucose, a central energy molecule, is present in all three nutritionally important disaccharides, paired with different partners. This explains why glucose intake rises whenever sucrose, maltose, or lactose is consumed. Which of the following correctly matches glucose with its disaccharide partner?
- Glucose + Galactose → Maltose
 - Glucose + Galactose → Sucrose
 - Glucose + Glucose → Sucrose
 - Glucose + Galactose → Lactose

The correct answer is: Glucose + Galactose → Lactose

11. During nutrition counseling, a nurse explains to a patient that while the body can store both glycogen and fat as energy reserves, the capacity for each type of storage differs greatly. Which of the following statements correctly explains this difference?
- Glycogen and fat are stored in equal amounts, but glycogen is preferred because it yields more energy per gram.
 - Fat storage is limited by the amount of dietary fat consumed, whereas glycogen can accumulate continuously without restriction.
 - Glycogen storage is limited because it holds water and occupies space in the liver and muscles, while fat can be compacted and stored in almost unlimited quantities in adipose tissue.
 - Both glycogen and fat are stored in unlimited quantities because they are compact and anhydrous energy reserves.

The correct answer is: Glycogen storage is limited because it holds water and occupies space in the liver and muscles, while fat can be compacted and stored in almost unlimited quantities in adipose tissue.

12. In a particular adult population, the Estimated Average Requirement (EAR) for vitamin B₆ is 1.1 mg/day. Based on this information, which of the following statements is most correct?
- There will also be a Recommended Dietary Allowance (RDA) set for this nutrient.
 - Neither an RDA nor an AI can be determined if only the EAR is known.
 - An EAR indicates that the nutrient poses a health risk at usual intake levels.
 - There will be an Adequate Intake (AI) set instead of an RDA.

The correct answer is: There will also be a Recommended Dietary Allowance (RDA) set for this nutrient.

13. In nutrition, energy in food is commonly expressed in kilocalories (kcal). Which of the following best defines one kilocalorie?
- The amount of energy required to raise the temperature of 1 gram of water by 1°C
 - The amount of energy stored in 1 gram of carbohydrate
 - The amount of energy required to raise the temperature of 1 kilogram of water by 1°C
 - The total energy the body uses in one hour of rest

The correct answer is: The amount of energy required to raise the temperature of 1 kilogram of water by 1°C

14. In nutritional assessment in clinical or hospital settings, nurses (or dietitians) may ask patients or use trained personnel to weigh every food and beverage consumed (and any leftovers or plate waste), record details of preparation methods and portion weight (often immediately before and after eating), and use established food composition tables to calculate nutrient intake precisely. This method is often used when very accurate intake data are needed (for example, in critical care, metabolic studies, or nutritional interventions) but requires high patient cooperation and can itself alter eating behavior. Which method is being described?
- Weighed food record
 - Dietary history
 - Estimated food record
 - Food frequency questionnaire

The correct answer is: Weighed food record

15. In the ongoing landmark Framingham Heart Study, which commenced in 1948 and in which the children and grandchildren of the original participants are still being followed today, what is the seminal finding of this study?
- Consumption of whole grains is associated with a healthy weight

- b. High blood cholesterol causes heart disease
 - c. Consumption of whole grains reduces the risk of high blood pressure
16. An investigator studying long-standing dietary patterns across southern Europe reports a habitual pattern marked by: predominant use of a monounsaturated culinary oil, plentiful seasonal fruits and vegetables, cereals (mostly unrefined) as staple carbohydrates, moderate fish and dairy intake, and relatively low consumption of red and processed meats. Which named dietary pattern does this description most closely represent?
- a. Traditional East Asian dietary pattern
 - b. Traditional Mediterranean dietary pattern
 - c. Western (industrialized) dietary pattern
 - d. DASH (Dietary Approaches to Stop Hypertension)

The correct answer is: Traditional Mediterranean dietary pattern

17. What is the primary storage form of glucose in humans, characterized by its highly branched polymeric structure that allows for rapid mobilization when energy demands increase?
- a. Amylose
 - b. Starch
 - c. Cellulose
 - d. Glycogen

The correct answer is: Glycogen

18. Kofi decides to replace refined foods such as white rice and pastries with whole grains, fruits, and vegetables rich in dietary fibre. Over time, this change is most likely to help reduce his risk of which health conditions?
- a. Iron deficiency and anemia
 - b. Rickets and scurvy
 - c. Obesity and diabetes
 - d. Dehydration and constipation

The correct answer is: Obesity and diabetes

19. Kojo is trying to eat healthier and wonders whether he should switch from white bread to whole grain bread. White bread is made from refined flour, where parts of the grain are removed during milling. Which key nutrients are lost when bran and germ are stripped away?
- a. Fat-soluble vitamins (A, D, E, K)
 - b. Fiber, B-vitamins, and minerals
 - c. Protein and water
 - d. Mostly starch only

The correct answer is: Fiber, B-vitamins, and minerals

20. Mrs. Asare purchased a packaged whole grain cereal from the local grocery store. While placing items in her reusable shopping bag, the nutrition facts label on the cereal box was partially torn, obscuring some critical information. Upon arriving home, she examined the label and found that per 100g serving, the cereal contains 1.4 mg of iron, but the percentage of the Guideline Daily Amount (%GDA) for iron was torn off and illegible. Fortunately, the footnote at the bottom of the label remained intact and indicated that the Guideline Daily Amount for iron used by the manufacturer is 14 mg. Based on this information, what percentage value was most likely printed in the torn area of the label for the %GDA of iron?
- a. 10%
 - b. 15%
 - c. 20%
 - d. 5%

The correct answer is: 10%

21. A nurse is counseling a patient about dietary fat intake and needs to explain the differences between various types of fatty acids to help the patient make informed food choices. Which of the following statements correctly distinguishes between saturated and unsaturated fatty acids in terms of their chemical structure, physical state at room temperature, and food sources?
- a. Both saturated and unsaturated fatty acids contain the same number of carbon-carbon double bonds, but saturated fats come from animal sources while unsaturated fats come exclusively from plant sources. Tropical oils like palm oil are unsaturated and have been proven to lower blood cholesterol, while products like margarine contain saturated fats from hydrogenated vegetable oils.
 - b. Saturated fatty acids are always liquid at room temperature regardless of their structure, with examples including palm oil and canola oil. Unsaturated fatty acids contain double bonds and are always solid at room temperature, such as those found in margarine and shortening, which are made from vegetable fats and are beneficial for blood cholesterol levels.
 - c. Saturated fatty acids have no carbon-carbon double bonds in their structure and tend to be solid at room temperature, with examples including stearic acid found in chocolate, meat, and butter. Unsaturated fatty acids contain one or more double bonds and are typically liquid at room temperature, such as oleic acid in olive oil, avocado, and fish oils. However, tropical oils like palm oil are exceptions, remaining liquid at room temperature despite being high in saturated fat, and palm oil has been proven to raise blood cholesterol.
 - d. Saturated fatty acids contain one or more carbon-carbon double bonds and tend to be liquid at room temperature, with examples including stearic acid in chocolate and butter. Unsaturated fatty acids have no double bonds in their carbon chains and are typically solid at room temperature, such as oleic acid found in olive oil and avocado.

The correct answer is: C

22. A nurse is reviewing food labels with a patient who has been advised to reduce saturated fat intake. The patient is confused about why products labeled as "made with vegetable oils" such as margarine, shortening, and commercially prepared pastries are often high in saturated fats. Which explanation correctly describes the food processing technique responsible for this?
- a. Hydrogenation is a process where hydrogen is added to the carbon double bonds of unsaturated fatty acids in vegetable oils, making them more saturated and converting them from liquid to semi-solid fats. This process increases shelf life and stability, which is why margarine, shortening, and commercially prepared products are high in saturated fats despite using vegetable oils.
 - b. Vegetable oils undergo a cooling process that changes their structure from unsaturated to saturated fats, making them solid at room temperature and suitable for use in margarine and commercial baking without any chemical modification.
 - c. Food manufacturers remove the unsaturated fatty acids from vegetable oils and replace them with saturated fats from animal sources to extend shelf life. This is why products made with vegetable fats behave like saturated fats.
 - d. Vegetable oils naturally contain high amounts of saturated fats, so when they are used in processed foods, the saturated fat content automatically increases. No additional processing is needed to make these products shelf-stable.

The correct answer is: A

23. A nurse is teaching a patient with type 2 diabetes how to evaluate whether their daily diet is balanced according to the Acceptable Macronutrient Distribution Ranges (AMDR). The patient reports that their typical daily intake provides a total of 2,000 kilocalories, with approximately 250 g of carbohydrate, 70 g of protein, and 80 g of fat. Based on the AMDR, which of the following best describes the patient's diet?
- a. Carbohydrates are within the AMDR, protein is within the AMDR, and fat is above the AMDR.
 - b. Carbohydrates are within the AMDR, protein is within the AMDR, and fat is within the AMDR.
 - c. Carbohydrates are below the AMDR, protein is within the AMDR, and fat is above the AMDR.
 - d. Carbohydrates are within the AMDR, protein is below the AMDR, and fat is above the AMDR.

The correct answer is: Carbohydrates are within the AMDR, protein is within the AMDR, and fat is above the AMDR.

24. Nurse Jennifer is conducting a nutrition assessment for her new patient, Augustine, a 42-year-old warehouse supervisor. To accurately estimate his Energy Requirements (EER), which combination of factors must she consider?
- a. Weight, height, medical diagnosis, and family history
 - b. Age, gender, weight, height, and physical activity level ✓
 - c. Gender, occupation, medication list, and appetite level

- d. Age, gender, weight, and dietary restrictions

The correct answer is: Age, gender, weight, height, and physical activity level ✓

25. Nurse Kwame is teaching a diabetes education class about the essential nutrients the body needs to function properly. He explains that all foods contain combinations of six classes of nutrients that serve different roles in the body. Which option correctly lists all six classes of nutrients?
- a. Carbohydrates, proteins, fats, vitamins, minerals, and water
 - b. Fruits, vegetables, grains, protein, dairy, and fats
 - c. Sugars, starches, proteins, fats, vitamins, and water
 - d. Carbohydrates, proteins, fats, vitamins, minerals, and fiber

The correct answer is: Carbohydrates, proteins, fats, vitamins, minerals, and water

26. Which of the following foods is classified as a whole grain rather than a refined grain?
- a. Brown rice
 - b. White rice (polished)
 - c. Fine maize flour used for tuo zaafi
 - d. White bread

The correct answer is: Brown rice

27. Which of the following statements best reflects the concept of malnutrition as understood in human nutrition?
- a. Individuals consuming energy-dense diets high in fats and sugars, resulting in obesity and metabolic disorders.
 - b. An adult population with widespread vitamin D and iron deficiencies despite adequate calorie consumption.
 - c. A condition encompassing nutrient deficiencies, excesses, or imbalances that adversely affect health.
 - d. A community where children suffer growth failure because of inadequate dietary energy and protein intake.

The correct answer is: A condition encompassing nutrient deficiencies, excesses, or imbalances that adversely affect health.

28. Which of these can be classified as energy nutrients I. Protein II. Carbohydrates III. Lipids IV. Water V. Vitamins and Minerals
- a. II, III, IV and V only
 - b. II only
 - c. I and II only
 - d. I, II and III only

The correct answer is: I, II and III only

29. Which of these is a non-nutrient component of food
- a. Lipids

- b. Carbohydrates
- c. Cholesterol
- d. Cobalamin

The correct answer is: Cholesterol

30. Which of these is an inorganic food substance
- a. Sterols
 - b. Sucrose
 - c. Potassium
 - d. Amino acids

The correct answer is: Potassium

31. A patient's lab results show low sodium and chloride levels. From which mineral category do these belong?
- a. Vitamins
 - b. Trace minerals
 - c. Water-soluble nutrients
 - d. Macro minerals

The correct answer is: Macro minerals

32. A person has been on a diet consisting mostly of packaged food, dried grains, and preserved meats for several months. They start to experience extreme fatigue, have bleeding gums and find that cuts and bruises are taking a very long time to heal. Based on the symptoms and diet described, what is the most likely nutrient deficiency?
- a. B-complex vitamins
 - b. Triglycerides
 - c. Vitamin A
 - d. Ascorbic acid

The correct answer is: Ascorbic acid

33. Phospholipids can interact with both water-based digestive fluids and fat-soluble nutrients in the small intestine. Which structural feature accounts for this water solubility?
- a. The shortened fatty acid chain length
 - b. The presence of two fatty acid chains instead of three
 - c. The glycerol backbone that holds the molecule together
 - d. The phosphate group coupled with a nitrogen-containing group

The correct answer is: The phosphate group coupled with a nitrogen-containing group

34. A plate of fried rice with chicken and vegetables, accompanied by a small glass of wine and a glass of water, was subjected to chemical analysis. The analysis revealed the following composition: 72 g of carbohydrates, 28 g of protein, 18 g of fat, and 12 g of alcohol. The meal also provided 200 mL of water, along with Vitamin C (55 mg), Iron

(14 mg), and Calcium (450 mg). Based on this information, calculate the total energy value of the meal.

- a. $\approx$ 646 kcal (Calories)
- b. $\approx$ 612 kcal (Calories)
- c. $\approx$ 849 kcal (Calories)
- d. $\approx$ 682 kcal (Calories)

The correct answer is: $\approx$ 646 kcal (Calories)

35. A scientist explains to a group of adults that while vitamin D is essential for strong bones, taking it in very high amounts can be harmful. In fact, regularly consuming more than 4,000 IU per day may cause kidney damage due to excessive calcium in the blood. The intake level described here represents which Dietary Reference Intake (DRI)?
- a. Recommended Dietary Allowance
 - b. Tolerable Upper Intake Level (UL)
 - c. Estimated Average Requirement (EAR)
 - d. Adequate Intake (AI)

The correct answer is: Tolerable Upper Intake Level (UL)

36. Scientists conducted a study on the vitamin C needs of school children in a town. After collecting data, they calculated that an intake of 40 mg per day would be enough to meet the needs of about half of the children in that population. This value represents which of the following Dietary Reference Intakes (DRIs)?
- a. Adequate Intake (AI)
 - b. Tolerable Upper Intake Level (UL)
 - c. Recommended Dietary Allowance (RDA)
 - d. Estimated Average Requirement (EAR)

The correct answer is: Estimated Average Requirement (EAR)

37. Sugar alcohols such as xylitol provide fewer kilocalories per gram compared to table sugar. What is the main reason for this reduced caloric contribution?
- a. They increase metabolic rate and promote faster energy expenditure.
 - b. They are incompletely digested and absorbed in the gastrointestinal tract.
 - c. They are fully converted into glycogen rather than glucose.
 - d. They are excreted unchanged in the urine without being metabolized.

The correct answer is: They are incompletely digested and absorbed in the gastrointestinal tract.

38. The traditional Mediterranean diet relies mainly on olive oil as its source of visible fat. The major fatty acid in olive oil contains one double bond in its carbon chain. Which type of fat does this describe?
- a. Saturated fat
 - b. Trans fat
 - c. Polyunsaturated fat